

Integration of Medical Imaging Systems Final Exam Solutions

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1 Final Exam

1.1 Instructions

1. Any text or figures beyond the page limit will not be given credit.
2. You may not use any of the figures in the lecture slides for your answers to any of the questions.
3. Figures must be developed and constructed on your own, preferably in PPT. Failure to include a figure will be grounds for point deductions.

Note: Please include Appendix for grading. The figures and tables otherwise do not fit.

1.2 Problem 1

Given an academic institution with the following four hospital sites:

Site 1: A large 1,000 bed Trauma Level 4 hospital with 6 CR Units, 4 Multi-Slice CT Scanners, 2 MRI Scanners, and 3 Angio Rooms.

Site 2: A cancer hospital with 100 beds with 1 CR, 1 Multi-Slice CT Scanner, and 1 MR Scanner.

Site 3: A 150 bed hospital with NO Emergency Department with 3 CR Units, 2 Multi-Slice CT Scanners, 2 MRI Scanners, and 1 Angio Room.

Site 4: An outpatient imaging center with 2 CR Units, 1 Multi-Slice CT Scanner, 1 MRI Scanner, and 1 Angio Room.

Please fully complete the following four steps with valid references. Four-page limit.

1. STEP 1: Design and describe a PACS solution for EACH site. Please include at least one figure of your own showing all the PACS and their corresponding components (as well as the total number for each of the PACS components that require multiple instances) involved. Please make sure you thoroughly describe your figures.
2. STEP 2: Design and describe an enterprise PACS solution that will support a Radiology Department with Radiologists that will read cases from ALL FOUR sites (eg, Data Storage and Backup Solutions, Type of PACS architecture, etc.).
3. STEP 3: Design a Network Infrastructure diagram that will support this Enterprise PACS for all four sites. Please be as specific as possible, including details of the following: 1) bandwidth estimations; 2) connectivity; 3) physical media and locations; and 4) hardware. You MUST include your own figure and make sure you describe it thoroughly.
4. STEP 4: Identify Single Points of Failure (SPOF) within the environment and provide possible Fault-Tolerant Solutions for the SPOF's.

1.2.1 Solution

1.2.2 STEP 1: Site-level PACS solution for EACH site

1.2.2.1 Design goals. Each site PACS design must: (i) accept DICOM storage (C-STORE) from modalities, (ii) ensure correct patient/study identity via RIS/ADT integration, (iii) provide local short-term cache for clinical continuity, (iv) support enterprise distribution to radiologists across all sites, and (v) satisfy security and audit requirements.

1.2.2.2 Core PACS components.

- **Acquisition / DICOM Router (1 per site):** modality ingress, routing rules, retries, and queueing.
- **RIS/ADT Interface (1 per site, logical):** HL7 v2 ADT/ORM messages update RIS scheduling state, which is exposed to modalities via a DICOM Modality Worklist (MWL) SCP, consistent with IHE Scheduled Workflow.
- **Local Cache / Edge Archive (Site 1 and Site 3 required; Site 2/4 optional):** short-term storage for continuity during WAN outages.
- **QC/Tech Workstations:** for technologist QA; scale with modality volume (Site 1 highest).
- **Radiologist Viewing (enterprise viewers preferred):** minimize local thick-client sprawl.

1.2.2.3 Series completion and workflow integrity. End-of-series detection at the gateway is governed by a combination of time-out logic, image number tracking, and Modality Performed Procedure Step (MPPS) notifications, enabling reliable series completion even when images arrive out of order. MPPS updates allow RIS and PACS to synchronize acquisition state, ensure UID integrity, and support downstream billing, dose tracking, and clinical metadata consistency.

1.2.2.4 Image acquisition gateway internals. Consistent with standard DICOM gateway architecture, each site-level ingest router comprises: (i) a local database management system (DBMS) maintaining non-persistent tables for patient, study, series, image, and queue state; (ii) a Storage Service Class Provider (SCP) that receives C-STORE objects from modalities; (iii) a Storage Service Class User (SCU) that forwards completed studies to the enterprise ingest tier; and (iv) short-term local image storage to support queueing, retries, and error recovery. Local gateway databases are non-authoritative and serve operational workflow tracking rather than long-term persistence.

1.2.2.5 Per-site sizing. Please see Table 1. To satisfy the requirement that each site list the total number of PACS components requiring multiple instances, Table 1 provides an explicit sizing baseline. Counts are conservative and should be validated against local throughput and staffing. Note that Site 1 uses redundant ingest and cache nodes (separate failure domains) due to highest throughput. Sites 2–3 benefit from a small edge cache to accelerate priors and reduce WAN dependency; Site 4 may omit cache if WAN reliability and enterprise latency requirements are strong.

1.2.2.6 Per-site instantiation
Site 1 (1,000-bed trauma; highest throughput). Deploy redundant DICOM router/queue nodes (active/standby or active/active) in separate failure domains, a larger local edge cache, dedicated modality VLAN(s), and the highest number of QC/tech workstations. Ensure rapid routing to enterprise for cross-site reads and trauma-priority workflow.

Site 2 (oncology; moderate volume, high priors). Use a single site router/queue with enterprise-grade

retry/queueing; deploy an edge cache to accelerate priors and avoid WAN-latency penalties for follow-up comparisons. Maintain strict identity verification via ADT/ORM + MWL integration to reduce demographic errors.

Site 3 (150-bed; no ED; multi-modality). Use a site router/queue sized for CT/MR throughput and a modest edge cache for continuity and priors. Emphasize scheduled workflows and cross-site distribution to centralized radiologists.

Site 4 (outpatient; smallest site). Use a single site router/queue with robust retransmit/queue behavior; edge cache is optional if WAN reliability is high. Prioritize secure WAN transport and rapid enterprise availability over local storage footprint.

1.2.2.7 Original Figure (site PACS pattern). Please see Figure 3. Figure 3 shows the site PACS pattern; Table 1 instantiates the pattern for each site with explicit instance counts (routers, caches, workstations, and HA network edges) to satisfy the component-total requirement.

1.2.3 STEP 2: Enterprise PACS supporting radiologists reading from ALL sites

1.2.3.1 Enterprise architecture. We adopt a hub-and-spoke enterprise model with centralized reading and lifecycle management:

- **Enterprise VNA / Archive (central):** authoritative long-term storage, lifecycle, retention, and cross-site priors.
- **Enterprise PACS Index + Workflow:** study indexing, worklist orchestration, rules (e.g., trauma priority).
- **Universal Viewer (zero-footprint):** reads across all sites with role-based access.
- **Disaster Recovery (DR) Archive:** separate failure domain (different AZ/region or data center).

- **Security + Audit:** centralized logging/SIEM feeds; least-privilege access.

1.2.3.2 Enterprise storage, backup, and lifecycle (required specifics).

- **Authoritative archive (VNA/object store):** tiered storage with hot cache for recent studies, warm object storage for active priors, and cold/archival tier for long-term retention.
- **Backup strategy:** immutable backups with scheduled restore tests; backups stored in a separate administrative and failure domain from the primary archive.
- **DR posture:** warm-standby archive in a separate region/site with defined RPO/RTO targets (stated explicitly in the RFP response).

Authoritative vs cache. The enterprise VNA/object store is the authoritative source of record; any site edge cache and viewer cache are non-authoritative accelerators and may be safely repopulated from the VNA after failures.

Retention (policy-bound). Imaging retention is enforced in the VNA via lifecycle rules (hot/warm/cold tiers) per institutional policy and regulatory requirements; audit logs and restore-test records are retained to support compliance.

1.2.3.3 Cross-site priors and worklist orchestration (required specifics).

- **Priors policy:** on exam scheduling (ORM/MWL), trigger prefetch of relevant priors from the VNA to the enterprise cache/viewer so that cross-site reads do not block on WAN latency.
- **Worklists:** rules-based worklists (e.g., trauma priority from Site 1) with role-based queues and audit logging for assignment, opening, and finalization events.

1.2.3.4 Original Figure (enterprise PACS). Please see Figure 4. Link labels in Figure 5 reflect the per-site $B_{\text{WAN,design}}$ targets in Table 2.

1.2.4 STEP 3: Network infrastructure diagram and bandwidth estimation

1.2.4.1 Connectivity & physical media.

- **Site 1 (hub-grade):** dual diverse fiber uplinks to enterprise (MPLS or SD-WAN over dual ISPs), redundant edge routers, redundant firewalls.
- **Sites 2–4:** dual-path WAN (primary fiber + secondary broadband/5G) with automatic failover.
- **Segmentation:** separate VLANs for modalities, clinical workstations, and server/edge appliances; enforce ACLs.
- **Encryption:** IPsec tunnels or private MPLS with additional app-layer controls as required.

1.2.4.2 Bandwidth sizing model. Let B_{WAN} be required WAN throughput for a site. Let D_{day} be expected imaging data volume per day (bytes/day) that must reach the enterprise, and let $\alpha \in (0, 1]$ be the fraction of a day available for transfers (e.g., $\alpha = 0.5$ if peak transfers must complete within 12 hours). Then,

$$B_{\text{WAN}} \geq \frac{8D_{\text{day}}}{86400\alpha} \quad (\text{bits/s}).$$

Recommended to add a safety factor $\gamma \geq 1.3$ to cover protocol overhead, retries, etc.

Proof. Over a time window $T = \alpha \cdot 86400$ seconds, transmitting D_{day} bytes requires moving $8D_{\text{day}}$ bits. Any link with constant capacity B_{WAN} can transmit at most $B_{\text{WAN}}T$ bits over that window. Feasibility demands $B_{\text{WAN}}T \geq 8D_{\text{day}}$, hence $B_{\text{WAN}} \geq \frac{8D_{\text{day}}}{T} = \frac{8D_{\text{day}}}{86400\alpha}$. A multiplicative safety factor γ preserves the inequality under overhead, yielding $B_{\text{WAN,design}} = \gamma B_{\text{WAN}}$. Thus, the design throughput requirement is

$$B_{\text{WAN,design}} = \gamma \cdot \frac{8D_{\text{day}}}{86400\alpha}.$$

□

1.2.4.3 Bandwidth inputs and per-site throughput targets (required estimations). Please see Table 2. To operationalize (1) bandwidth estimation, we state explicit daily volume assumptions D_{day} and compute $B_{\text{WAN,design}} = \gamma \cdot \frac{8D_{\text{day}}}{86400\alpha}$. Replace the assumed D_{day} values with measured data (modality logs, historical study sizes, and compression policy). The numeric circuit tiers must follow from the computed $B_{\text{WAN,design}}$.

1.2.4.4 Physical media, locations, and connectivity.

- **Site 1:** dual diverse fiber entrances to the MDF, separate carriers, separate demarcation points; SD-WAN/MPLS termination in a locked network room; redundant power (UPS).
- **Sites 2–4:** primary fiber/cable to MD-F/IDF, secondary broadband or 5G at a separate demarc with automatic failover via SD-WAN.
- **Segmentation:** separate VLANs for modalities, clinical workstations, and PACS/edge appliances; enforce ACLs at L3 boundary and firewall.

1.2.4.5 Network hardware.

- **Per site:** HA firewall pair, dual WAN edge routers (or SD-WAN appliances), and redundant L3 switching (stack/M-LAG) in MDF; modalities and QC stations connect via access switches in IDFs.
- **Enterprise core:** redundant SD-WAN/MPLS hub termination, load balancers for ingest/index, and separate management network for monitoring/logging.

1.2.4.6 Original Figure (network).

Please see Figure 5.

1.2.5 STEP 4: ID SPOFs & propose fault-tolerant solutions

1.2.5.1 Common SPOFs.

- **Single DICOM ingest/router at a site.**

- **Single WAN circuit or single edge firewall/router.**
- **Single enterprise ingest endpoint or single metadata/index service.**
- **Single enterprise storage array or single availability zone.**
- **Single identity/authentication provider without failover.**
- Single HL7 interface engine / integration broker (ADT/ORM/ORU) without HA.
- Single enterprise database for PACS index/workflow without clustering/failover.
- Single DNS/NTP/time source (certificate validation and audit correlation failure risk).
- Single PKI/certificate authority path (expired certs can halt DICOMweb/viewer access).
- Single power/UPS path for site MDF or enterprise ingest stack.

1.2.5.2 Fault-tolerant solutions. Please see Table 3.

- Redundant site ingest: active/standby DICOM routers with shared queues or hot failover.
- Dual WAN + diverse paths: separate carriers/entrances; automatic failover (SD-WAN).
- Enterprise high availability: ingest and index services behind load balancers; multi-node clusters.
- Storage resilience: replicated object storage or mirrored arrays; immutable backups; regular restore tests.
- DR posture: warm standby in separate failure domain with defined RPO/RTO targets.

1.2.5.3 Data integrity invariant.

Throughout the ingest and archival pipeline, PACS enforces an “always two copies” rule: images are not deleted from a modality, gateway, or RAID tier until downstream persistence is explicitly verified, preventing data loss during transient failures or partial outages. Distinct failure domains are defined as independence across at least:

- (i) power/UPS, (ii) rack/host, (iii) network path/carrier, and (iv) identity/certificate services.

1.2.5.4 SPOF elimination claim.

Proof. Model clinical delivery as a directed dependency graph $G = (V, E)$ with sources S (site ingest endpoints) and sink t (enterprise viewer). A vertex v is a single-vertex SPOF for reachability to t if v dominates t with respect to some $s \in S$, i.e., every directed path from s to t contains v . To remove such a SPOF, deploy two replicas v_1, v_2 in distinct failure domains and rewire dependencies so that each predecessor of v can reach at least one of $\{v_1, v_2\}$ and at least one of $\{v_1, v_2\}$ can reach each successor of v . Under a single-component failure model (at most one vertex fails at a time), removing either v_1 or v_2 leaves the other replica available; thus a directed path from each $s \in S$ to t remains. Therefore, no single replica dominates t after redundancy is added, eliminating single-vertex SPOFs. Residual risks arise from shared failure domains or multi-failure events, handled via DR and operational controls. $\square$

1.2.5.5 Acceptance testing and operational validation. Prior to clinical go-live, the PACS environment undergoes structured acceptance testing covering end-to-end workflows (RIS order $\rightarrow$ acquisition $\rightarrow$ archive $\rightarrow$ distribution $\rightarrow$ display), back-end functionality (ingest, indexing, priors retrieval), dynamic performance under concurrent load, and simulated failure scenarios. Testing explicitly verifies redundancy claims by exercising gateway failover, WAN failover, archive unavailability, and restore procedures, ensuring the absence of unmitigated single points of failure before clinical deployment.

Note: Archive/VNA denotes the authoritative repository; gateways provide local DICOM routing/buffering; viewers are non-authoritative diagnostic clients.

1.3 Problem 2

Acceptance Testing is crucial for the implementation of a PACS in the clinical environment. It is comprised of system technical test scenarios to insure that the PACS can operate correctly both during normal operations and downtime situations. Using the PACS configuration example shown in the figure attached below, along with the PACS characteristics listed directly after the figure.

Please develop and describe all three steps with valid references. Three-page limit.

1. Define a **REALISTIC** dataset of PACS Image Studies (eg, What Modality Types would you include, Number of Images in each Study, etc) to be used for the Acceptance Test for performance and image quality assessment. Explain your reasoning.
2. Using the textbook and lecture notes provided, design technical test scenarios to verify functionality of each PACS component. Identify the points of failure for the figure below and include them within the technical test scenarios. Please make sure the technical test scenarios include detailed instructions of what you plan to do during the Acceptance Test.
3. Design a technical test scenario to verify if the Contingency Plan (**RED** and **BOLD** text) shown below in the figure and listed in the PACS characteristics will work for the main archive server. Again, include detailed step-by-step instructions.

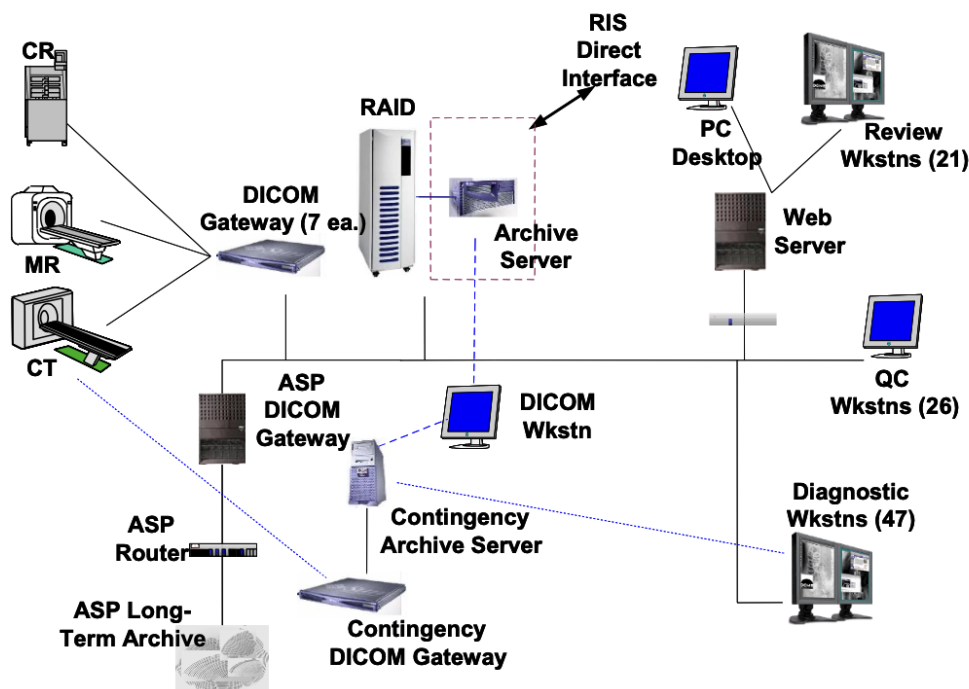


Figure 1

The following PACS characteristics are applied to Figure 1.

1. Client-Server Architecture
2. Multiple DICOM Acquisition Gateways
3. RIS/PACS Interface is Brokerless
4. RAID has Redundant Fiber Paths to Each Server
5. Web Server
6. Clinical Review Workstations Use Web App
7. QC & Diagnostic Workstations Use Client-Server App
8. ASP Offsite Archive is Long-Term Storage
9. ASP Offsite Archive has OC-3 Connectivity
10. Completely Separate Backup Server Used as Contingency Plan if Main Archive Server Goes Down
11. Any PACS Exams Stored in Backup Server During Downtime Are Exported Into Main Server via a standalone DICOM Workstation Afterwards

1.3.1 Solution

1.3.2 STEP 1: Define a realistic dataset of PACS image studies

1.3.2.1 Dataset design principles. The acceptance dataset must be (i) clinically representative across modalities and exam complexity; (ii) large enough to stress ingest, indexing, storage, and retrieval; (iii) controlled and labeled for objective pass/fail criteria; and (iv) safe to use (de-identified or synthetic identifiers, with audit logging enabled).

1.3.2.2 Recommended modality mix.

Figure 4 shows CR, CT, and MR feeding a DICOM Gateway into an Archive Server with RAID; downstream clients include Web-based clinical review workstations and client-server QC/Diagnostic workstations. A realistic dataset should therefore include:

- **CR:** adult chest and extremity radiographs (low size, high volume).
- **CT:** head CT (emergent), CT chest/abdomen/pelvis (high size, multi-series).
- **MR:** brain MRI (multi-sequence), spine MRI (multi-series).

1.3.2.3 Example dataset sizing. Let the acceptance dataset include the following minimum study counts:

- **CR:** $N_{\text{CR}} = 300$ studies, ~ 2 images/study (e.g., PA/LAT), $\approx 10\text{--}30$ MB/study.
- **CT:** $N_{\text{CT}} = 120$ studies, ~ 3 series/study, $\sim 300\text{--}1200$ images/study, $\approx 300\text{--}1500$ MB per study.
- **MR:** $N_{\text{MR}} = 80$ studies, $\sim 5\text{--}8$ series/study, $\approx 150\text{--}800$ MB per study.

Total dataset volume is typically in the range of $\sim 80\text{--}250$ GB depending on protocol variability (slice count, bit depth, series count, and compression policy); acceptance testing uses ideal fixed representatives selected from within each range. This size is sufficient to exercise ingestion concurrency, indexing, caching, RAID throughput, web viewing, and client-server diagnostic workflows without requiring multi-day transfers.

1.3.2.4 Metadata realism. Populate (or verify) standard DICOM tags needed for workflow and routing: Patient ID, Accession Number, Study/Series UUIDs, Modality, Study Date/Time, Performing/Requesting Physician, Body Part Examined, and Institution Name. Use test patient cohorts stratified by modality and exam type. Ensure de-identification is consistent with institutional policies and regulations.

1.3.2.5 Performance sampling requirement (for objective metrics). For each major workflow (ingest, query/retrieve, display launch, export/reconciliation), measure latency/throughput over repeated trials to obtain stable estimates.

Proof. Let $X_1, \dots, X_n$ be i.i.d. latency measurements with mean μ and finite variance σ^2 . By the central limit theorem, for large n , the standardized mean

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

is approximately standard normal. Thus an approximate $(1 - \delta)$ two-sided confidence interval for μ has half-width

$$\varepsilon \approx z_{1-\delta/2} \frac{\sigma}{\sqrt{n}},$$

where $z_{1-\delta/2}$ is the $(1 - \delta/2)$ quantile of $\mathcal{N}(0, 1)$. Solving for n yields the sufficient condition

$$n \geq \left(\frac{z_{1-\delta/2} \sigma}{\varepsilon} \right)^2.$$

In practice, σ is unknown; if $\hat{\sigma}$ is obtained from a pilot sample, one may replace $z_{1-\delta/2}$ by the corresponding Student- t quantile $t_{1-\delta/2, n-1}$ or inflate n by a safety factor, maintaining coverage under estimation error. $\square$

1.3.3 STEP 2: Technical test scenarios verifying each PACS component

1.3.3.1 Scope. We design scenarios to verify functionality and failure handling for:

DICOM acquisition gateways, brokerless RIS/PACS interface, archive server, RAID with redundant fiber paths, web server, clinical review workstations (web app), QC/-diagnostic workstations (client-server app), ASP offsite long-term archive (OC-3), and operational monitoring/audit.

1.3.3.2 Global acceptance criteria.

- **Correctness:** study completeness (all instances received), correct demographics, correct study/series counts, and correct routing.
- **Availability:** services remain operational under expected load; graceful degradation under injected faults.
- **Performance:** meets locally defined thresholds for ingest throughput, query latency, and time-to-display for representative studies.
- **Security/Audit:** authentication enforced, access logged, and least-privilege roles verified.

Archive receipt verification includes Storage Commitment-style confirmation that all transmitted instances are durably persisted before source deletion. Acceptance testing is considered successful only when all scenarios pass without unresolved critical failures and when recovery tests demonstrate zero data loss under injected faults. DICOM Storage Commitment (N-ACTION / N-EVENT-REPORT) or an equivalently verifiable vendor mechanism.

1.3.3.3 Figure 1 points-of-failure map.

Key single-component and interface failure points implied by Figure 1 include: (i) DICOM Gateway(s): AE Title/routing misconfiguration, queue overflow, NIC or host failure; (ii) RIS Direct Interface: HL7 connectivity loss, field-mapping errors, ADT merge inconsistencies; (iii) Archive Server: service outage, index/database corruption, disk-full conditions; (iv) RAID + fiber paths: multipath misconfiguration, single-path loss, controller alarms and degraded-

mode throughput collapse; (v) Web Server: authentication/session failures, CPU/memory saturation under concurrent use; (vi) Review/QC/Diagnostic Workstations: client-server connectivity failures, permissions/-role misconfiguration, workstation performance issues; (vii) ASP Router/OC-3 link: link outage, firewall/VPN policy errors, bandwidth saturation; (viii) Contingency Archive/Gateway: failover routing failure, inability to query downtime studies, export-back duplication due to AE/UID handling. These failure points are exercised explicitly in Scenarios 2.1–2.7 and Scenario 3.1.

1.3.3.4 Scenario 2.1 — DICOM acquisition gateways (multiple gateways).

Goal: validate multi-gateway ingest, routing rules, queueing, and retry.

1. Configure at least two gateways with distinct AE Titles and verify network reachability (DICOM echo).
2. Send CR batch (e.g., 50 studies) concurrently through Gateway A and CT batch (e.g., 20 studies) through Gateway B using C-STORE.
3. Verify archive receipt: for each accession, confirm StudyInstanceUID present and expected instance counts match modality output.
4. Inject transient gateway outage: stop Gateway B mid-transfer; confirm queued resend upon restart and no data loss.

Likely failure points: AE Title misconfiguration, routing table errors, queue overflow, association timeouts, duplicate UID handling.

Expected results:

- All transmitted studies are received by the archive with complete instance counts matching modality output.
- Each study has a unique and valid StudyInstanceUID and is indexed exactly once.
- Studies interrupted by the injected gateway outage are successfully resent from the queue upon gateway restart.

- No studies are lost or duplicated as a result of the outage.

1.3.3.5 Scenario 2.2 — Brokerless RIS/PACS interface (HL7 + MWL).

Goal: validate that orders/ADT updates are reflected without an intermediary broker.

1. Create test orders in RIS (ORM) and ADT updates for patient merges/updates.
2. Verify MWL shows correct scheduled procedures and identifiers.
3. Acquire a study using MWL-selected patient; confirm demographics match RIS and accession is consistent end-to-end.
4. Perform ADT correction (e.g., name change/merge) and verify PACS reconciliation behaves as expected.

Likely failure points: HL7 connectivity, field mapping, duplicate MRNs, merge semantics, MWL filtering.

Expected results:

- Modality worklists accurately reflect RIS orders, including patient identifiers and scheduled procedures.
- Acquired studies match RIS demographics and accession numbers end-to-end.
- ADT corrections (e.g., name changes or merges) are reflected correctly in PACS without creating duplicate studies.

1.3.3.6 Scenario 2.3 — Archive server + RAID with redundant fiber paths.

Goal: validate storage integrity, redundancy, and path failover.

1. Ingest mixed dataset segment (e.g., 10 CT + 10 MR) and confirm successful commit to archive.
2. Simulate single fiber path failure (disable one path); verify I/O continues on alternate path without study loss.
3. Verify RAID status reports degraded/-failed path while service remains available.
4. Re-enable failed path; confirm resynchronization and return-to-normal status.

Likely failure points: multipath misconfig, RAID controller alarms ignored, performance collapse in degraded mode.

Expected results:

- Archive continues to accept and store studies during a single fiber path failure.
- No data corruption or study loss occurs while operating in degraded mode.
- RAID and storage systems correctly report degraded and recovered states.

1.3.3.7 Scenario 2.4 — Web server + clinical review workstations (web app).

Goal: validate web viewing, authentication, and typical clinical flows.

1. Authenticate as a clinical reviewer role; verify role-based access to permitted studies only.
2. Open representative CR and CT studies; record time-to-first-image and basic tools (window/level, zoom, scroll).
3. Verify web session timeout and re-authentication behavior.
4. Load test: simulate multiple concurrent web viewers (e.g., 20 sessions) to verify responsiveness.

Likely failure points: web server saturation, session/auth misconfiguration, cross-origin issues, viewer codec incompatibility.

Expected results:

- Authorized clinical users can authenticate and access only permitted studies.
- Representative CR and CT studies are successfully displayed with basic tools (window/level, zoom, scroll).
- Time-to-first-image and interactive responsiveness meet site-defined acceptance thresholds.
- User access and viewing actions are recorded in audit logs.

1.3.3.8 Scenario 2.5 — QC workstations (client-server app). **Goal:** validate technologist QC workflow and data correction steps.

1. Log in as QC role; verify ability to flag incomplete studies and request re-send.

2. Perform study-level QC: check series counts, orientation, and metadata fields.
3. Verify QC flags propagate to workflow/-worklists and audit logs record actions.

Likely failure points: client-server connectivity, permissions, inconsistent QC states.

Expected results:

- QC users can flag incomplete or incorrect studies.
- QC status changes are reflected in PACS workflow and visible to downstream users.
- All QC actions are logged for audit purposes.

1.3.3.9 Scenario 2.6 — Diagnostic workstations (client-server app). Goal: validate radiologist-grade performance and hanging protocols.

1. Log in as diagnostic role; verify advanced tools (MPR for CT if supported, key images, measurements).
2. Retrieve priors; confirm correct patient matching and fast access (cache effectiveness if present).
3. Validate report workflow linkage to RIS (as applicable) and audit trail.

Likely failure points: prior matching errors, cache invalidation, workstation GPU/-driver issues, latency sensitivity.

Expected results:

- Diagnostic users can access advanced viewing tools appropriate to modality.
- Prior studies are correctly matched to the patient and retrievable.
- Retrieval and interaction performance meets site-defined diagnostic workstation thresholds.
- Diagnostic access and actions are logged.

1.3.3.10 Scenario 2.7 — ASP offsite long-term archive (OC-3). Goal: validate export to long-term storage and retrieval path.

1. Export a labeled subset (e.g., 10 CT, 10 MR) to ASP archive over the OC-3 link.
2. Confirm successful receipt at ASP and retention tagging (long-term).

3. Request retrieval of exported studies and verify integrity (UIDs, instance counts) and acceptable retrieval latency.

Likely failure points: link saturation, firewall/VPN policies, checksum/integrity failure, incomplete exports.

Expected results:

- Selected studies are successfully exported to the ASP archive with intact UIDs and instance counts.
- Exported studies are retrievable from the ASP archive without data corruption.
- Retrieval latency consistent with site-defined expectations: long-term storage.

1.3.4 STEP 3: Contingency plan test for main archive server downtime (RED/BOLD plan)

1.3.4.1 Contingency plan requirement (from characteristics). During main archive server downtime, studies are stored in a completely separate backup/-contingency server. After recovery, exams stored during downtime are exported back into the main server via a standalone DICOM workstation.

1.3.4.2 Scenario 3.1 — Planned failover test (end-to-end). Goal: verify continuity of operations and post-downtime reconciliation exactly as specified.

1. **Pre-checks:** confirm baseline ingest to main archive is healthy; record service status and logs.
2. **Initiate downtime:** stop main archive services (or isolate network path) in a controlled window.
3. **Verify detection:** confirm DICOM gateways detect failure and route/store to contingency archive server.
4. **Ingest during downtime:** transmit a labeled downtime batch:

$$\mathcal{D}_{\text{down}} = \{10 \text{ CR}, 5 \text{ CT}, 3 \text{ MR}\}$$

using distinct accessions (tagged “DOWN_TEST”).

5. **Verify clinical access:** from clinical review web app and diagnostic workstation, confirm studies in $\mathcal{D}_{\text{down}}$ are searchable/viewable via the contingency path, with correct access control and audit logs.
6. **Recover main archive:** restore main archive services; confirm health and availability.
7. **Standalone export:** using the standalone DICOM workstation, export all studies from contingency to main archive (C-MOVE/C-STORE as supported).
8. **Reconciliation checks:** verify that each study in $\mathcal{D}_{\text{down}}$ is present in the main archive with matching StudyInstanceUID and full instance counts; verify no duplicates and correct indexing.
9. **Post-checks:** confirm that contingency retains (or purges) data according to policy after successful export; verify logs/audit trails for the entire event.

1.3.4.3 Pass/fail criteria (contingency).

- **Continuity:** all $\mathcal{D}_{\text{down}}$ studies are accepted during downtime with no loss.
- **Usability:** web and diagnostic clients can retrieve/view downtime studies with correct role-based controls.
- **Reconciliation:** after restoration, 100% of $\mathcal{D}_{\text{down}}$ is present on main archive with exact instance counts; no duplicate studies created.
- **Auditability:** all failover, ingest, access, export, and recovery actions are logged.

1.3.4.4 Failure points to explicitly test (must be observed or injected).

- Gateway routing fails to switch to contingency.
- Contingency archive accepts studies but index/view path cannot query them.
- Standalone DICOM workstation export produces duplicates (UID collisions or misconfigured AE Titles).
- Post-export indexing lag prevents timely visibility on main archive.

Expected results:

- During main archive downtime, all studies are accepted by the contingency archive without loss.
- Studies ingested during downtime are accessible to authorized clinical and diagnostic users.
- After restoration, all downtime studies are present on the main archive with matching StudyInstanceUIDs and instance counts.
- No duplicate studies are created during export-back.
- All failover, access, export, and recovery actions are recorded in audit logs.

1.3.4.5 System of record. The central archive/VNA in the metro data center is the sole legal system of record for imaging data and associated metadata. Local caches, gateways, and streamed representations are non-authoritative and exist only to support transport resilience and diagnostic efficiency.

1.3.4.6 Compression policy by use case.

- **Archival storage:** uncompressed or loss-less compression only.
- **WAN transport (constrained links):** diagnostically acceptable compression for time-sensitive reads.
- **Viewer streaming:** derivative representations (e.g., JPEG 2000 / WADO-RS) generated on demand.

1.3.4.7 WAN degradation or outage.

The hub-and-spoke architecture allows additional imaging centers or modalities to be onboarded by deploying an additional gateway and WAN endpoint, without changes to expert-center workflows or archival governance. In the event of WAN degradation or outage, local gateways buffer studies and continue acquisition without data loss. Diagnostic workflows may temporarily rely on representative or streamed images for preliminary interpretation, with full-fidelity synchronization occurring once connectivity is restored. This approach prioritizes patient care continuity while preserving long-term data integrity.

1.4 Problem 3

Design a teleradiology system with identification AND location of the various systems required, including network connectivity for a Radiology group that consists of one expert center and three imaging centers of various sizes. PLEASE PROVIDE A FIGURE and make sure to explain it.

The goal is to digitally store and distribute all exams from each of the Imaging Centers together with a RIS. You can assume that the expert center and Imaging Centers 2 & 3 are in a major metropolitan area. However, Imaging Center 1 is in a rural and remote area which will impact what level of services are available (eg, High Speed Internet access, etc).

Develop and justify with valid references. Ensure to include figure. Three-page limit.

1. Expert Center: Primary Diagnosis only – no Modalities.
2. Imaging Center 1: Two conventional radiography systems for screening (no CR and no DR)
3. Imaging Center 2: One CR and one CT (Trauma Center).
4. Imaging Center 3: One CT and one MRI

Ensure to provide an assumption and explain your reasoning for the workload of each of the Imaging Centers.

Note that for the teleradiology solution, make sure to address the storage and distribution of images throughout the teleradiology system, where, legally, you must preserve the raw original size images within the archive while still saving space and supporting the need for compression for timely diagnosis in limited bandwidths. Please provide a data workflow for the image studies to support your case.

1.4.1 Solution

1.4.2 Assumptions and workload characterization

1.4.2.1 Geographic assumptions. The Expert Center and Imaging Centers 2 and 3 are located in a major metropolitan area with access to high-speed fiber connectivity. Imaging Center 1 is located in a rural area with limited and variable bandwidth.

1.4.2.2 Workload assumptions (daily averages). These assumptions are chosen to be clinically realistic and sufficient for sizing and justification.

- **Imaging Center 1 (Rural Screening):** Two conventional radiography rooms; ~ 80 screening exams/day, low image count per study. Since Imaging Center 1 uses conventional radiography (no CR/DR), images are digitized via an approved film/plate digitizer or capture system before DICOM transmission through the local gateway.
- **Imaging Center 2 (Trauma):** One CR and one CT; ~ 40 CR studies/day and ~ 30 CT studies/day, with time-sensitive reads.
- **Imaging Center 3 (Advanced Imaging):** One CT and one MRI; ~ 25 CT studies/day and ~ 20 MRI studies/day, large multi-series datasets.
- **Expert Center:** No modalities; diagnostic interpretation only, serving all three imaging centers.

1.4.2.3 Image size characteristics. Typical uncompressed (or lossless-compressed) study sizes are assumed as:

- Radiography: ~ 10–30 MB/study
- CT: ~ 300–1500 MB/study
- MRI: ~ 200–800 MB/study

1.4.3 System architecture overview

1.4.3.1 Core design principle. All original image data are preserved at full fidelity (raw or lossless-compressed) within a central archive. For diagnostic distribution, appropriate compression and caching strategies

are applied to support timely interpretation over constrained networks, especially for the rural site.

1.4.3.2 Major system components.

- **Local Acquisition Systems:** modality consoles at each imaging center.
- **Local DICOM Gateway (each imaging center):** buffering, routing, and retry for WAN transmission.
- **Central Enterprise Archive/VNA (metro):** authoritative storage of all original image data.
- **RIS (central):** order management, worklists, and reporting integration.
- **Expert Center Diagnostic Workstations:** primary interpretation only.
- **Edge Cache (Imaging Center 1):** temporary local cache for resilience and delayed synchronization.

1.4.4 Original Figure: Teleradiology system architecture

Please see Figure 6.

1.4.5 Data storage and legal preservation requirements

1.4.5.1 Original image preservation.

All studies are stored in the central archive in their original resolution and bit depth, using either uncompressed storage or lossless compression. This satisfies medico-legal requirements for image retention duration, immutability of originals, auditability of access, and traceable linkage between derived and original objects.

1.4.5.2 Compression strategy for distribution. To support timely diagnosis over limited bandwidth:

- **Lossless compression** is used for archival storage and when full-fidelity transfer is feasible.
- **Visually lossless or diagnostically acceptable compression** (e.g., JPEG 2000) is used for streaming/remote viewing, particularly from Imaging Center 1.

1.4.5.3 Justification. The original images remain preserved in the archive, while compressed representations are used only for transport and display. This separation ensures legal compliance without sacrificing diagnostic timeliness.

1.4.6 Data workflow (end-to-end)

1.4.6.1 Normal operation.

1. Images are acquired at an imaging center and sent to the local DICOM gateway.
2. The gateway transmits studies over the secure WAN to the central archive/VNA.
3. The RIS updates worklists and notifies the Expert Center of available studies.
4. Expert radiologists retrieve studies; for large datasets or rural-origin studies, images are streamed or transmitted using compression as appropriate.
5. Final reports are generated at the Expert Center and returned to the RIS.

1.4.6.2 Raw-preservation vs. transport-optimized derivatives. For each acquired study, the system maintains (i) an authoritative original-fidelity object in the central archive (uncompressed or lossless-compressed), and (ii) an optional transport/view representation for constrained links, via shared `StudyInstanceUID` and explicit derivation metadata. At the rural site, the DICOM gateway may transmit an initial diagnostically usable compressed/streamed representation for timely reads, simultaneously scheduling off-peak synchronization of the full-fidelity study to the central archive. The central archive remains the legal system of record; any derived representations are non-authoritative and are linked to the same patient/study identifiers for traceability.

1.4.6.3 Rural bandwidth accommodation. For Imaging Center 1:

- Studies are buffered locally if WAN bandwidth is insufficient.
- Priority or representative images may be transmitted first for rapid preliminary interpretation.

- Full-fidelity studies synchronize to the central archive during off-peak hours.

1.4.7 Performance and scalability considerations

1.4.7.1 Latency sensitivity. Trauma CT studies from Imaging Center 2 are prioritized for rapid transmission and interpretation. Screening radiography from Imaging Center 1 tolerates delayed full synchronization. This prioritization is enforced via RIS-driven urgency flags and gateway routing rules (priority queues) for trauma CT.

1.4.7.2 Scalability. The centralized archive and RIS allow additional imaging centers or modalities to be added without altering the Expert Center workflow, supporting future growth.

1.5 Problem 4

Figure 2 will be used for the entirety of Problem 4. Please develop and justify all parts with valid references. Three-page limit total.

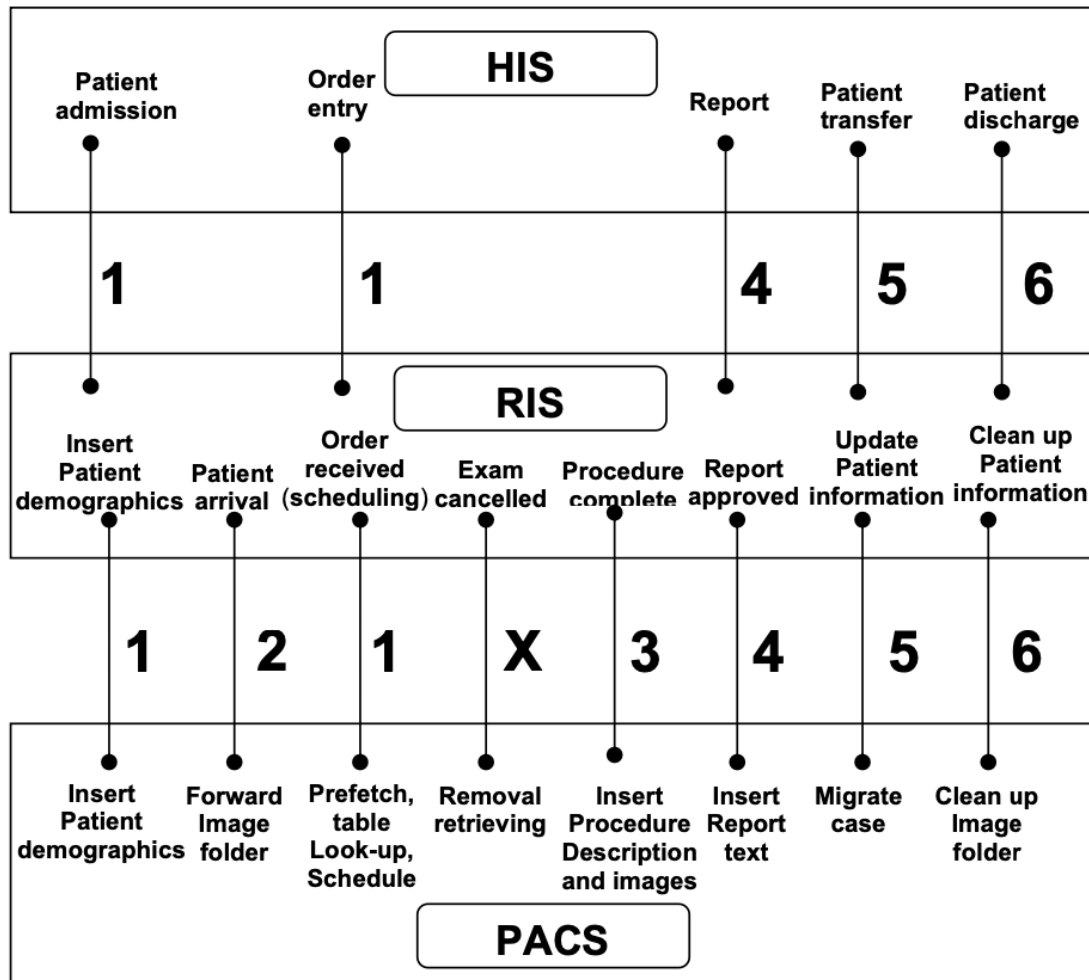


Figure 2

1.5.1 Part I

1. Please indicate the Radiology Clinical Workflow that corresponds to each of the numbered steps.
2. Please indicate what number the steps marked × should be and put arrows in the proper direction for each of the integration workflow steps to indicate the data flow. Make sure to explain your reasoning thoroughly.

1.5.1.1 Solution

1.5.1.2 (1) Radiology clinical workflow corresponding to each numbered step. We interpret Figure 2 as the standard radiology enterprise workflow with synchronized HIS $\leftrightarrow$ RIS $\leftrightarrow$ PACS transactions:

1. **Step 1 (Registration & Ordering):** patient admission/registration and order entry (HIS), order received/scheduled (RIS), and PACS prefetch/schedule table updates.
2. **Step 2 (Arrival/Check-in):** patient arrival event (RIS) and PACS folder/-worklist preparation for acquisition.
3. **Step 3 (Acquisition/Procedure Complete):** imaging is performed; procedure completion (RIS) triggers PACS ingestion of procedure description and images.
4. **Step 4 (Interpretation/Reporting):** report is approved (RIS) and finalized report content is stored/distributed (HIS, PACS).
5. **Step 5 (Patient transfer/merge updates):** transfer/ADT updates propagate; RIS updates patient information; PACS migrates/merges cases as required.
6. **Step 6 (Discharge/Closure):** discharge triggers cleanup: RIS closes/cleans patient info; PACS cleans study/image folders per policy.

1.5.1.3 Step-to-system event mapping (auditable). To avoid vendor-specific assumptions, we define each step by (i) an observable trigger, (ii) a standards-based message/object class, and (iii) verifiable evidence in HIS/RIS/PACS (UIs, databases, audit logs, or DICOM query endpoints). See Table 6.

1.5.1.4 Interface contracts (explicit standards + join keys). We make the interface contracts implementable by stating the minimum identifiers used to bind transactions across systems (order $\rightarrow$ acquisition $\rightarrow$ archive $\rightarrow$ report).

- **Cross-system join keys (minimum):** MRN (patient), visit/encounter ID (if used), accession number (order/exam), procedure code(s), scheduled date/time, and later StudyInstanceUID (imaging object identity).
- **Step 1 (Registration & Ordering):** HIS $\rightarrow$ RIS uses HL7 v2 ADT (patient/visit) and ORM (order). *Requirement:* RIS must generate/accept an accession and persist the MRN $\leftrightarrow$ accession link.
- **Step 2 (Arrival/Check-in and alternative cancel branch):** MWL must reflect scheduled procedure steps keyed by accession/procedure identifiers. *Requirement:* modalities shall not acquire against an accession that is cancelled/invalid per RIS policy.
- **Step 3 (Acquisition/Procedure Complete):** modality $\rightarrow$ PACS uses DICOM Storage (C-STORE) and must preserve StudyInstanceUID consistency within a study. *Requirement:* PACS/VNA stores objects and indexes them under accession/MRN linkage.
- **Step 4 (Interpretation/Reporting):** RIS $\rightarrow$ HIS uses HL7 v2 ORU for the finalized report. If PACS stores report artifacts, they must be linked to the same accession/study identifiers (*policy-dependent*).
- **Step 5 (Transfer/Merge Updates):** HL7 v2 ADT updates/merge semantics update demographics and identifiers. *Requirement:* RIS/PACS reconciliation must not create duplicate patient records for the same MRN.
- **Step 6 (Discharge/Closure):** cleanup affects transient work artifacts (folders, caches) but does not delete authoritative imaging, which is preserved under retention policy.

1.5.1.5 (2) Determine the step number for the $\times$ markers and add arrows for data flow.

1.5.1.6 Claim: The steps marked × correspond to Step 2 (an alternative branch at the Arrival/Check-in stage).

In Figure 6, the × appears at Exam cancelled (RIS) and Removal retrieving (PACS). Assuming a linearized clinical state machine with ordered transitions as depicted in Figure 2, operationally, cancellation is a pre-acquisition state transition that occurs after scheduling (Step 1) but before procedure completion (Step 3). In the same time window, the site either (i) registers arrival (Step 2) or (ii) cancels/no-shows (Step 2 alternative). Therefore, the × should be labeled 2.

1.5.1.7 Arrow semantics (data objects and triggers). In Figure 7, each arrow corresponds to a specific class of message/object:

- **HIS→RIS:** HL7 v2 ADT/ORM messages (patient identity, encounters, orders).
- **RIS→PACS:** scheduling/procedure context (worklist/encounter linkage) used to bind accession and patient/study identifiers.
- **Modalities→PACS:** DICOM instances via C-STORE, keyed by StudyInstanceUID and accession number.
- **RIS→HIS (report):** HL7 v2 ORU final report (or status + impression) as the authoritative clinical communication.
- **PACS/RIS→Audit:** security events (logins, access, reconciliation, cancellation cleanup) as immutable audit records.

1.5.1.8 Expected outcomes for the × cancellation branch (acceptance criteria, not vendor folklore). Let the × events denote Exam cancelled (RIS) and Removal retrieving (PACS), occurring after Step 1 and before Step 3. Because vendor behavior varies, we specify outcomes as testable criteria with explicit evidence sources.

- **C1 (Order state consistency):** *Requirement:* the order/exam shall be

marked cancelled in the system-of-record (HIS or RIS, per site policy). *Evidence:* status field in HIS/RIS UI or database; cancellation timestamp + user/service. *Pass:* status = Cancelled (or Discontinued) and is queryable by accession.

- **C2 (RIS scheduling closure):** *Requirement:* the scheduled procedure step associated with the accession shall be closed/cancelled with a reason. *Evidence:* RIS exam/procedure record; cancellation reason code. *Pass:* cancellation is stored and appears in the RIS audit trail.
- **C3 (Modality worklist safety):** *Requirement:* the modality worklist shall not present the cancelled SPS as available for acquisition (either removed or clearly indicated as cancelled, depending on vendor). *Evidence:* MWL query results (screenshot/logs) before vs after cancellation. *Pass:* cancelled accession cannot be selected for a new acquisition without an override workflow.
- **C4 (PACS transient artifact handling):** *Requirement:* transient prefetch/schedule artifacts associated with the cancelled accession shall not remain in an “active acquisition” state. *Evidence:* PACS scheduler/prefetch queue status; workfolder/cache listing; system logs. *Pass:* artifacts are removed, invalidated, or marked terminal-cancelled per policy, and do not trigger downstream reads.
- **C5 (Auditability):** *Requirement:* the cancellation event and any cleanup actions shall be logged with user, time, and identifiers. *Evidence:* RIS/PACS audit logs (exportable) referencing MRN/accession and event type. *Pass:* auditor can reconstruct the cancellation timeline end-to-end.

Recall the structural timing argument; operational evidence enforced by criteria C1–C5.

Continued on the next page.

Proof. Let the radiology workflow be modeled as a state machine with ordered clinical states

Ordered $\rightarrow$ Arrived $\rightarrow$ Completed $\rightarrow$ Reported
 $\Rightarrow$ Transferred $\rightarrow$ Discharged,

corresponding to steps 1, 2, 3, 4, 5, 6, respectively. A cancellation event transitions from Ordered to a terminal state Cancelled and, by definition, occurs prior to Completed. Therefore the cancellation transition occurs strictly after step 1 and strictly before step 3. The unique integer step label satisfying $1 < k < 3$ is $k = 2$, so the $\times$ marker corresponds to step 2 as an alternative branch between steps 1 and 3. $\square$

1.5.1.9 Integration data-flow arrows (HIS/RIS/PACS). Figure 7 provides an original redraw with arrow directions indicating the operational data flow. (HIS $\rightarrow$ RIS uses HL7 ADT/ORM/ORU; RIS $\rightarrow$ PACS uses scheduling/worklist and procedure status; modalities $\rightarrow$ PACS uses DICOM C-STORE; reporting propagates back to HIS and into PACS as a report object/text depending on implementation.) Please see Figure 7 and Figure 5.

1.5.2 Part II

Assuming the HIS/RIS/PACS are now fully integrated, design the following ePR based on the clinical application of Minimally Invasive Spinal Surgery (MISS) procedures as discussed in the lecture by answering the following questions

1. What specific input images/studies (eg, imaging modality and body region) and patient related information should be acquired for MISS patients ONLY? (Remember, not EVERY PACS imaging study is needed so be very specific).
2. Design and develop a component diagram of the ePR system and include a figure OF YOUR OWN. Please include a storage system, a web-server system, and a client GUI (eg, WHAT data will be presented?).
3. System integration to any existing clinical imaging and information systems that will provide the input data listed in item 1 (eg, How will the system connect to the hospital clinical systems). Please include this in your figure in Problem 4 Part II Question 2.
4. Implementation of the ePR system within the clinical environment and identify which user groups are to be trained (eg, WHO will use WHAT).

1.5.2.1 Solution

1.5.2.2 Context. Assuming HIS/RIS/-PACS are fully integrated, we design an ePR focused on MISS patients only. The design restricts data acquisition to what is clinically required for MISS planning, intra-operative decision support, and post-operative follow-up, while preserving authoritative imaging in PACS/VNA.

1.5.2.3 (1) Specific input images/studies and patient information for MISS patients ONLY

1.5.2.4 Required imaging inputs (MISS-focused). We include only studies that directly support MISS planning and outcome tracking:

- **Lumbar/Thoracic spine MRI (pre-op):** sagittal and axial sequences sufficient for neural compression and soft-tissue assessment (e.g., T1/T2-weighted). Body region: affected spinal levels.
- **Spine CT (pre-op, as indicated):** high-resolution bony anatomy for instrumentation planning (pedicle/lamina morphology), particularly when hardware placement is anticipated.
- **Standing spine radiographs (pre-op/post-op):** alignment/instability assessment when clinically indicated (e.g., flexion/extension or upright AP/lateral depending on local protocol).
- **Intra-operative fluoroscopy (or O-arm/CBCT if used):** limited set of key images confirming level localization and instrumentation positioning.
- **Post-op targeted imaging (as indicated):** limited radiographs/CT for hardware check or complication workup; avoid broad imaging not tied to MISS follow-up.

1.5.2.5 Required patient-related information (MISS-only).

- **Identifiers and scheduling:** MRN, encounter, procedure code, operative level(s), laterality, surgeon, date/time.

- **Clinical indications:** diagnosis (e.g., stenosis, disc herniation), symptoms, neurologic findings, red flags.
- **Pre-op risk and constraints:** allergies, anticoagulation, relevant comorbidities, implants/contraindications to MRI.
- **Operative metadata:** planned approach, levels treated, instrumentation plan, intra-op events.
- **Outcome tracking:** pain/function scores (site-defined instruments), complications, re-admissions, re-operations.

1.5.2.6 MISS-only scoping rules (explicit inclusions/exclusions). To satisfy the requirement that not every PACS study is needed, we enforce the following MISS-specific scope policy:

- **Include only level-targeted spine studies** relevant to the operative plan (e.g., L4–L5) and immediate peri-operative follow-up.
- **Exclude unrelated imaging** (e.g., non-spine CT chest/abdomen/pelvis, unrelated extremity MRI, historical spine studies not used for current planning).
- **Timing rule (example policy):** pre-op MRI/CT must be within an institution-defined recency window (e.g., 3–6 months) unless new neurologic deficit/red-flag symptoms justify repeat imaging.
- **Post-op imaging is symptom- or protocol-triggered only** (hardware check or complication workup); routine broad surveillance imaging is excluded.

1.5.2.7 MISS minimum data elements (required fields for completeness). Each MISS case in the ePR must store the following required structured fields (or the record is flagged incomplete):

- **Operative targeting:** spinal level(s), laterality, planned approach, procedure code.
- **Clinical baseline:** pre-op neurologic status summary, primary indication, red-flag checklist (binary fields).

- **Contraindications/constraints:** MRI contraindications, anticoagulation status, implant constraints if revision.
- **Outcome tracking:** selected pain/function instrument(s), complications, readmissions, re-operations.

1.5.2.8 (2) ePR component diagram (original figure) with storage, web server, client GUI Please see Figure 8.

1.5.2.9 Design principle. The ePR stores structured MISS summaries and pointers to authoritative imaging in PACS/VNA. Images are viewed through a secure viewer using DICOMweb/WADO-RS or a PACS-integrated viewer, rather than duplicating full-fidelity imaging into the ePR.

1.5.2.10 Client GUI: required screens and what is presented . The MISS ePR client must present, at minimum, the following data views (each is testable):

1. **MISS Dashboard:** patient identifiers (MRN/encounter), indication, operative level(s) or laterality, schedule, and status.
2. **Planning Summary:** structured plan fields (approach, instrumentation plan, key risks/constraints).
3. **Key Imaging Launch:** one-click viewer launch into the correct StudyInstanceUID/series (contextual deep link).
4. **Intra-op Checklist:** level localization confirmation + implant/instrument meta-data entry (minimal required fields).
5. **Post-op Follow-up:** outcome score collection + complication flags + targeted imaging links (if obtained).

1.5.2.11 Integration contracts (explicit data objects per interface). To make Figure 8 implementable, we require the following interface contracts:

- **HIS/EMR → ePR:** ADT (demographics/encounters) and clinical context (FHIR resources, if available).
- **RIS → ePR:** orders/scheduling and finalized report metadata (ORM/ORU or

site-equivalent).

- **PACS/VNA → ePR viewer:** DICOMweb query/retrieve (QIDO-RS/WADO-RS); ePR stores immutable identifiers `StudyInstanceUID`, accession, timestamps for auditability.

1.5.2.12 Access control and authorization enforcement (RBAC/SSO).

Authentication is handled via institutional SSO/identity provider (IdP). Authorization is enforced at **both** (i) the ePR application/API (role-based permissions for MISS workflows) and (ii) the imaging viewer (token-based authorization and patient-context checks). Viewer launch from the ePR passes a time-limited authorization context; all accesses are logged for audit.

1.5.2.13 (3) System integration to existing clinical imaging/information systems

1.5.2.14 System-of-record hierarchy (conflict resolution). To prevent data divergence, we define authoritative sources as follows:

- **HIS/EMR:** system of record for patient identity, encounters, and admissions/discharges/transfers (ADT).
- **RIS:** system of record for orders, scheduling, procedure state, and final radiology report text/status.
- **PACS/VNA:** system of record for imaging objects (full-fidelity DICOM) and imaging metadata identifiers (e.g., `StudyInstanceUID`).
- **MISS ePR:** stores derived MISS summaries, structured care-path data, and immutable pointers to authoritative imaging/report objects; it does not override HIS/RIS/PACS truth.

1.5.2.15 Integration interfaces.

- **HIS/EMR → ePR:** ADT (demographics/encounters) and clinical context via HL7 v2 or FHIR resources.
- **RIS → ePR:** order and scheduling con-

text via ORM; finalized procedure/report context via ORU (or site-equivalent).

- **PACS/VNA → ePR viewer:** imaging access via DICOMweb (WADO-RS/QIDO-RS/STOW-RS) or enterprise viewer integration; ePR stores immutable references (StudyInstanceUID, Accession, timestamps).

1.5.2.16 Traceability requirement (immutable linkage). Each MISS ePR record MUST store immutable identifiers linking to authoritative systems: MRN, encounter ID, procedure code, accession number, and StudyInstanceUID. Any derived views (thumbnails/streams) are treated as non-authoritative and must be referentially bound to the same identifiers. This ensures end-to-end auditability and prevents clinical divergence between ePR summaries and the imaging system of record.

1.5.2.17 Clinical implementation plan (deployment + rollout + downtime). Implementation within the clinical environment must specify:

- **Deployment location:** hospital network zone for ePR web server and database (on-prem or approved private cloud), with segmentation from general-purpose networks.
- **Pilot rollout:** start with one MISS service line and defined user cohort; expand after acceptance criteria are met.
- **Downtime procedure:** if integration is unavailable, clinicians can (i) access PACS directly and (ii) enter a minimal MISS note later; reconciliation rules define how data are re-synchronized.
- **Go-live support:** named on-call owner(s) for application, integration engine, and PACS viewer integration.

1.5.2.18 Training completion criteria (what “trained” means). A user group is considered trained only if they can complete their role-critical tasks end-to-end:

- **Surgeons:** open a MISS case, confirm level/laterality, launch key imaging, finalize planning summary.
- **OR staff:** complete localization confirmation workflow and implant/instrument metadata entry.
- **Radiology:** view MISS context, tag key images/series, and ensure report linkage is correct.
- **Clinic staff:** collect outcome scores and schedule follow-up tasks without duplicative data entry.
- **IT/Security:** verify RBAC policies, audit logs, and downtime/recovery steps.

1.5.2.19 (4) Implementation in clinical environment & user-group training

1.5.2.20 Training Matrix.

- **Surgeons:** MISS ePR dashboard, operative planning view, post-op follow-up module, imaging viewer launch.
- **OR staff (nursing/techs):** intra-op checklist, level localization confirmation workflow, implant/instrument metadata entry.
- **Radiologists:** imaging viewer workflows, MISS-specific structured impression fields, key-image tagging.
- **Clinic staff:** scheduling verification, outcome score collection, patient instructions and visit documentation.
- **IT/Security:** role-based access control, audit log review, integration monitoring, downtime procedures.

1.5.2.21 Operational acceptance criteria. The integrated MISS ePR deployment is acceptable if: (i) all role-based accesses are enforced (least privilege) and auditable; (ii) viewer launches from the ePR resolve to the correct patient/study within the PACS/VNA without manual re-search; (iii) procedure and report status updates propagate within institution-defined latency bounds; and (iv) downtime procedures preserve clinical continuity with reconciliation steps documented and tested.

2 Appendix

2.1 Problem 1: Step 1

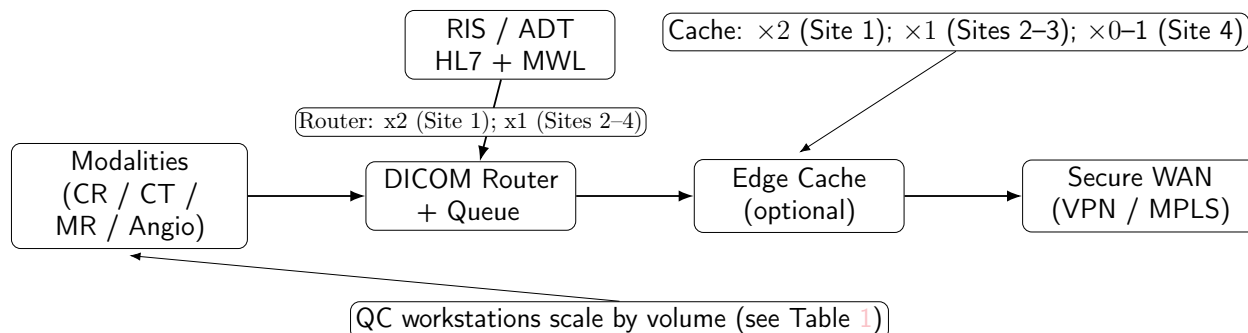


Figure 3: Site-level PACS pattern (logical) with sizing cues. Modalities transmit DICOM objects (C-STORE) to an ingress router with queuing and RIS-driven workflows (HL7/MWL); studies may land in a local edge cache to support continuity and priors before secure WAN transport to the enterprise environment. Concrete per-site instance counts are listed in Table 1.

Component	Site 1	Site 2	Site 3	Site 4
DICOM Router / Gateway (ingest + queue)	2 (A/S)	1	1	1
RIS/ADT interface endpoint(s) (logical)	1	1	1	1
Local cache / edge archive node(s)	2	1	1	0–1
QC / Tech workstation(s)	6	2	3	1
Radiologist diagnostic workstation(s) (local)	6–10	1–2	1–2	0–1
Enterprise (zero-footprint) viewer seats (local)	as needed	as needed	as needed	as needed
Site firewall(s) (HA)	2	2	2	2
Site edge router(s) (dual WAN termination)	2	2	2	2
MWL service endpoint (via RIS or broker)	1	1	1	1
Time sync + log forwarder (NTP/Syslog to SIEM)	1	1	1	1

Table 1: Minimum site-level PACS component instance counts (baseline). A/S denotes active/standby high availability. “0–1” indicates the component is optional and deployed if WAN reliability/latency requires local continuity and fast priors. Viewer seats have a minimum baseline of one per staffed reading position (scales with staffing).

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The central archive/VNA and RIS are protected by a DR/backup archive deployed in a separate failure domain (separate region and/or independent infrastructure). Resiliency targets are tiered by workload criticality, prioritizing diagnostic availability and data integrity for clinical reads over downstream analytical workloads. Specifically, RPO is ≤ 15 minutes for critical clinical services and ≤ 24 hours for non-critical analytics, while RTO for restoration of clinical reading workflows is ≤ 2 hours. These objectives are finalized in accordance with institutional policy and vendor-supported replication and failover capabilities; the complete resiliency rationale and assumptions are provided in Appendix ??.

2.2 Problem 1: Step 2

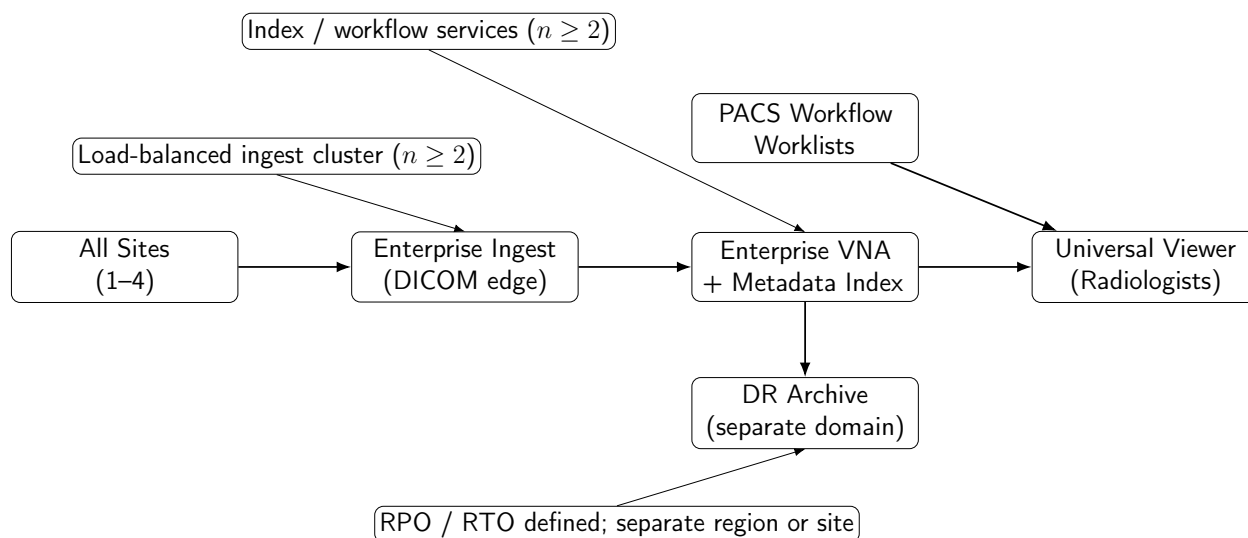


Figure 4: Enterprise PACS (hub-and-spoke). All sites deliver studies to a centralized ingest tier (load-balanced, $n \geq 2$) that feeds an authoritative VNA and metadata index; workflow services generate cross-site worklists; radiologists read via a universal viewer. A DR archive in a separate failure domain supports resilience with defined RPO/RTO objectives.

2.3 Problem 1: Step 3

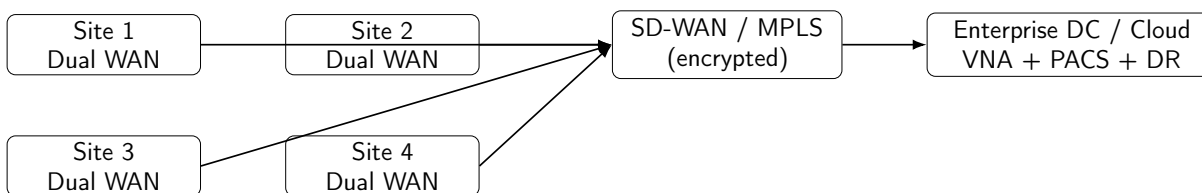


Figure 5: Network topology: all sites connect via encrypted SD-WAN/MPLS to the enterprise environment; dual WAN per site reduces outage risk.

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	Site 1	Site 2	Site 3	Site 4
Assumed D_{day} (TB/day)	3.0	0.4	0.8	0.2
Transfer window α	0.5	0.5	0.5	0.5
Safety factor γ	1.3	1.3	1.3	1.3
$B_{\text{WAN,design}}$ (Mbps)	722	96	193	48
Recommended primary circuit tier	1–2 Gbps	300–500 Mbps	500–1000 Mbps	200–300 Mbps

Table 2: Example WAN sizing inputs and design targets based on interpretation of D_{day} as the post-compression, post-routing imaging payload that must traverse the WAN, use 1 TB = 10^{12} bytes for sizing, and treat γ as an aggregate multiplier covering protocol overhead, retries, etc.

This bandwidth sizing establishes a necessary lower bound under constant-rate assumptions; final circuit tiers must additionally account for peak concurrency, backlog recovery after outages, and read-side traffic (priors and viewers).

2.4 Problem 1: Step 4

SPOF	Fault-tolerant mitigation
Single site router/ingest	Active/standby routers with shared queues; tested failover
Single WAN circuit	Dual diverse circuits + SD-WAN policy-based failover
Single enterprise ingest/index	Load balancer + multi-node cluster ($n \geq 2$)
Single storage array/AZ	Replication across failure domains + immutable backups + DR
Single IdP/PKI/DNS/NTP	HA services (redundant resolvers/time sources) + monitored cert lifecycle

Table 3: SPOF-to-mitigation traceability (minimum set).

2.5 Problem 3: Teleradiology System Architecture

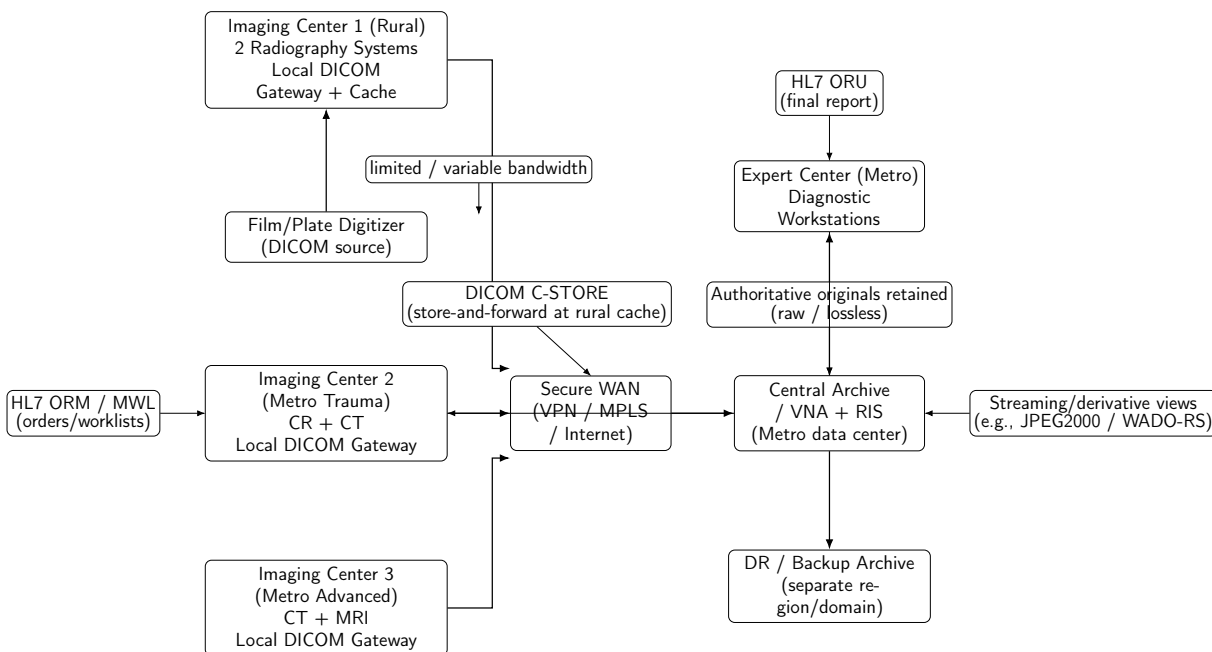


Figure 6: Teleradiology system architecture with explicit workflow cues. Imaging centers send studies via secure WAN to a central archive/VNA and RIS (metro). Original-fidelity images are preserved centrally; streaming/derivative views support timely expert diagnosis, including for the rural site with limited bandwidth.

2.5.0.1 Security and compliance considerations. All study transport between imaging centers and the metro data center should use encrypted channels (e.g., site-to-site VPN or SD-WAN with strong cryptography) and mutual authentication for DICOM and HL7 endpoints. Access to the central archive/VNA, RIS, and diagnostic viewer should follow least-privilege role-based access control, with centralized identity management and multi-factor authentication for remote/privileged users. Systems should maintain tamper-evident audit logs (study access, exports, report finalization), and PHI should be encrypted at rest on central and DR storage with appropriate key management. Operational controls should include segmentation of clinical networks, hardened viewer/workstation builds, patch management, backup/restore testing, and periodic security reviews to support HIPAA-aligned confidentiality, integrity, and availability requirements (as applicable to the deployment environment).

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Site	Workload assumption (daily avg.)	Volume (GB/day)
Imaging Center 1 (Rural)	~80 radiography exams/day; ~20 MB per study (typical)	1.6
Imaging Center 2 (Metro Trauma)	~40 CR/day @ ~20 MB; ~30 CT/day @ ~600 MB	18.8
Imaging Center 3 (Metro Advanced)	~25 CT/day @ ~600 MB; ~20 MRI/day @ ~400 MB	23.0

Table 4: Daily imaging workload and raw data volume assumptions used for downstream WAN and storage sizing. Assumes two rooms operating 8 hours/day at 5 exams/hour/room. These values inform, but do not directly determine, WAN circuit provisioning, which is derived from post-compression payloads and peak-behavior modeling. Please see Table 2.

2.5.0.2 Relation between Table 4 and WAN circuit sizing (Table 2). For WAN engineering, these raw volumes are converted to the post-compression, post-routing payload that must traverse the WAN, Table 4 estimates raw daily imaging volumes from workload assumptions. denoted D_{day} in Table 2. Operationally, this mapping can be expressed as

$$D_{\text{day}} \approx V_{\text{raw}} \cdot \eta_{\text{comp}} \cdot \eta_{\text{route}},$$

where V_{raw} is the raw volume (GB/day), $\eta_{\text{comp}} \in (0, 1]$ captures effective compression/packaging (lossless for archive vs. diagnostically acceptable for transport/streaming as applicable), and $\eta_{\text{route}} \geq 1$ accounts for routing overhead such as retries, retransmissions, and any required duplicate deliveries (e.g., local cache plus central archive). Table 2 then sizes the WAN to meet a transfer window α with safety factor γ , providing headroom for peak-hour bursts and backlog catch-up after outages; therefore circuit tiers are expected to exceed simple daily-average throughput.

2.6 Problem 4: Part I

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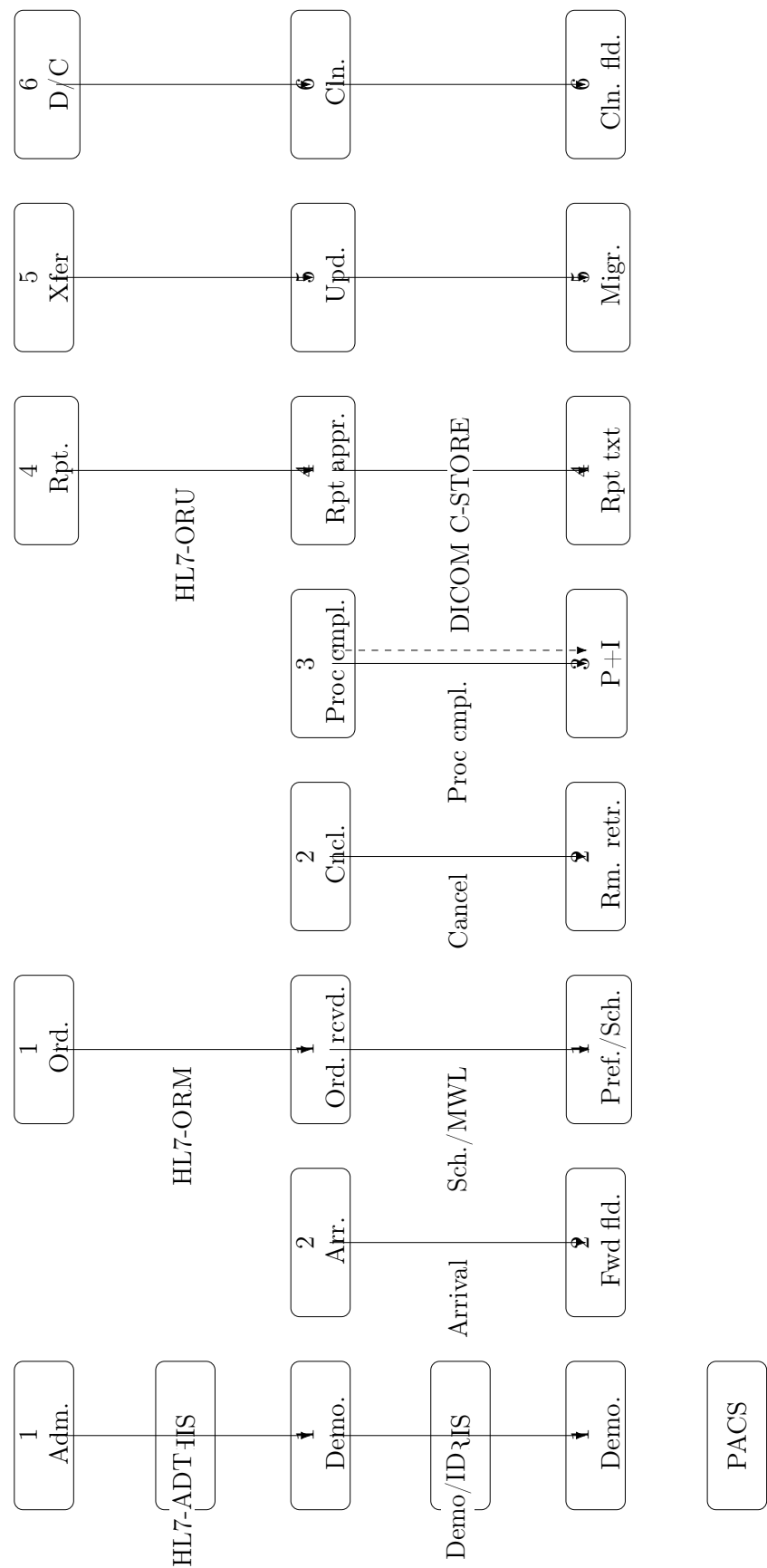


Figure 7: HIS/RIS/PACS integration workflow with message classes and transfer cues. Step 2 (exam cancellation / retrieval removal) represents an alternative branch between Step 1 (ordering) and Step 3 (procedure completion).

Shorthand	Meaning
Adm.	Admission
Ord. ent.	Order entry
Ins. demo.	Insert demographics
Pt arr.	Patient arrival / check-in
Ord. rcvd.	Order received
Exam cncl.	Exam cancelled
Proc cmpl.	Procedure complete
Rpt. / Rpt appr.	Report / report approved
Upd. info	Update information
Cln-up / Cln-up fld.	Cleanup (folder cleanup in PACS context)
Fwd fld.	Forward folder
Pref./Sch.	Prefetch / schedule
Rm. retr.	Remove retrieving
Ins. proc+img.	Insert procedure + images
Ins. rpt txt	Insert report text
Migr. case	Migrate case
HL7-ADT	HL7 v2 ADT message class (admit/discharge/transfer; demographics)
HL7-ORM	HL7 v2 ORM message class (orders/scheduling)
HL7-ORU	HL7 v2 ORU message class (results/reporting)
Sch./MWL	Scheduling context + DICOM Modality Worklist (MWL)
Demo/ID link	Demographics + identifier linkage (MRN/visit/accession binding)
Rpt ref.	Report reference / finalized report payload (implementation-dependent)
DICOM C-STORE	DICOM Storage Service (C-STORE) for image/object transfer

Table 5: Definitions of shorthand labels used in Figure 7.

Step	Trigger (clinical)	Interface object(s)	Minimum verification evidence (pass if present)
1	Registration + order placed	HL7 v2 ADT + ORM (order)	MRN/visit exists (HIS/RIS); accession/procedure scheduled (RIS); PACS prefetch/schedule entry created or updated (PACS scheduler/log)
2	Arrival/check-in <i>or</i> <i>cancellation/no-show branch</i>	MWL refresh; status update event(s)	MWL reflects correct SPS availability (or cancellation); RIS exam state updated; PACS workfolder/cache created (arrival) or removed/invalidated (cancel branch)
3	Acquisition + procedure complete	DICOM C-STORE (images); procedure complete status	DICOM objects stored (PACS/VNA); instance counts match modality output; procedure state = completed in RIS
4	Interpretation + final report	HL7 v2 ORU (results) and/or SR/encapsulated docs	Final report exists (RIS/HIS); PACS report object present if policy supports it; timestamps and authoring user recorded
5	Transfer/merge	HL7 v2 ADT updates/merge events	Patient identity updates propagated; accession/study ownership reconciled; PACS case migration event logged
6	Discharge/closure	Retention + cleanup policies	Encounter closed; transient folders cleaned (PACS); audit trail retained; authoritative imaging retained per policy

Table 6: Auditable system event mapping for the HIS/RIS/PACS integration workflow.

2.7 Problem 4: Part II

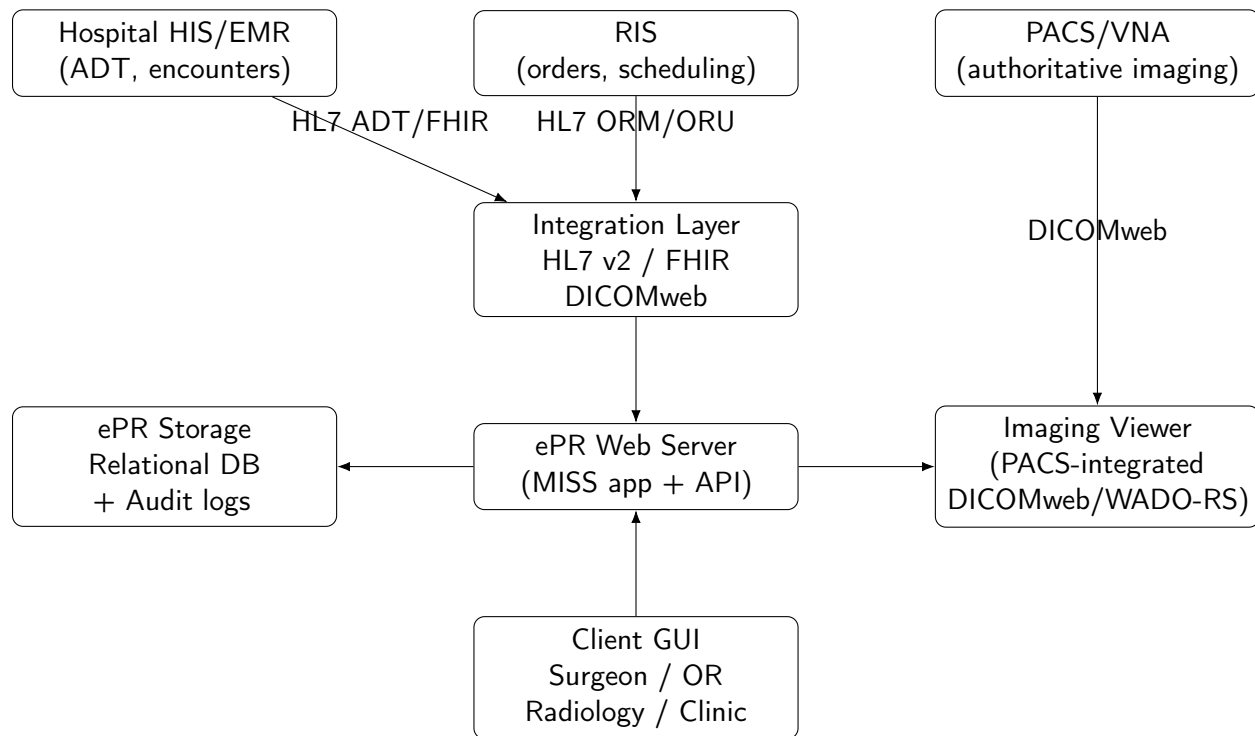


Figure 8: Original ePR component diagram for MISS. The ePR stores structured MISS data and audit logs; authoritative imaging remains in PACS/VNA and is accessed via a secure viewer using DICOMweb/WADO-RS (or equivalent), avoiding unnecessary duplication of full-fidelity images.

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